

IBVAP — Deployment & Production Setup

11. Deployment Architecture & Secret Management

1. Actual Production Deployment Pipeline

```
graph TD; A[Local Code Base / Google Colab] --> B[Git Source Control (GitHub Repository)]; B --> C[Streamlit Community Cloud (Public Web Host)]; C --> D[Cloud Database Integration (MongoDB Atlas / Supabase)];
```

2. Deployment Configurations & Files

- Main Web Entry Point: `src/app.py` or `streamlit_app/app.py`
- Dependencies Specification File: `streamlit_app/requirements.txt`
- `streamlit>=1.38.0`
- `pandas>=2.2.0`
- `plotly>=5.24.0`
- `supabase-py>=2.7.0`
- `Pillow>=10.4.0`
- `requests>=2.32.0`
- `pymongo>=4.6.0` (included in single-file setup)
- `certifi>=2024.0.0`
- UI Theme Configuration: `streamlit_app/.streamlit/config.toml`

3. Environment Variables & Secrets Management

To prevent hardcoding credentials in public GitHub repositories, IBVAP uses environment variable abstraction and Streamlit Secrets (`st.secrets`).

Configured Secret Keys

- `MONGODB_URI` / `MONGO_URI` / `MONGO_URL`
- `SUPABASE_URL` / `VITE_SUPABASE_URL`
- `SUPABASE_ANON_KEY` / `VITE_SUPABASE_ANON_KEY`

Access Code Snippet ([src/app.py](#) Lines 83–95)

```
def get_mongo_uri():
    for key in ["MONGODB_URI", "MONGO_URI", "MONGO_URL"]:
        uri = os.environ.get(key, "").strip()
        if uri:
            return uri
    try:
        if hasattr(st, "secrets"):
            for key in ["MONGODB_URI", "MONGO_URI", "MONGO_URL"]:
                if key in st.secrets:
                    return str(st.secrets[key]).strip()
    except Exception:
        pass
    return ""
```

4. Step-by-Step Deployment Guide

Local / Server Execution

```
# 1. Clone repository
git clone https://github.com/your-org/ibvap.git
cd ibvap

# 2. Install dependencies
pip install -r streamlit_app/requirements.txt
pip install ultralytics opencv-python pymongo certifi python-dotenv

# 3. Set environment variables (or create .env file)
export MONGODB_URI="your_mongodb_connection_string"

# 4. Launch Streamlit dashboard
streamlit run src/app.py
```

Streamlit Community Cloud Deployment

- Push codebase to GitHub.
- Log into [Streamlit Community Cloud](https://streamlit.io/cloud).
- Select repository and branch.
- Set Main file path to [src/app.py](#) or [streamlit_app/app.py](#).
- Under **Advanced Settings** $\rightarrow$ **Secrets**, paste secrets:

```
MONGODB_URI = "mongodb+srv://..."
SUPABASE_URL = "https://..."
SUPABASE_ANON_KEY = "eyJ..."
```

- Click **Deploy**. Public URL is generated automatically.